

Opgave 1 – Perslucht

a)

Bijvoorbeeld:

- Gemakkelijk rechtlijnige bewegingen te maken
- Geen retourleidingen
- Gemakkelijk te genereren en op te slaan
- Soepele leidingen
- Geen vervuiling bij lekkages

① PER JUIST VOORDEEL
 -1 PER FOET

2 P.

b)

Bij lucht-afname zal de werkdruk niet direct dalen.

1 P.

c)

$$30^{\circ}\text{C} \approx 28 \text{ g/m}^3$$

$$10^{\circ}\text{C} \approx 8 \text{ g/m}^3 \quad \textcircled{1}$$

$$30^{\circ}\text{C} \Rightarrow 28 \cdot 5 \cdot 0.6 = 84 \text{ g H}_2\text{O} \quad \textcircled{1}$$

$$10^{\circ}\text{C} \Rightarrow 8 \cdot 5 \cdot 1 = 40 \text{ g H}_2\text{O} \quad \textcircled{1}$$

$$\textcircled{1} 44 \text{ g H}_2\text{O op de bodem}$$

4 P.

d)

$$F = p \cdot A$$

$$A_1 \cdot p_1 = A_2 \cdot p_2 \Rightarrow p_2 = \frac{A_1 \cdot p_1}{A_2}$$

$$F = \frac{1 \times 10^5}{4} \cdot \frac{1}{4} \pi \cdot 0.2^2 = 3142 \text{ N}$$

① ①

$$p_2 = \frac{1 \cdot 1}{0.5} = 2$$

$$2 \text{ ABS} = 7_{\text{NCL}}$$

ALT. REDENATIE
 MAG OOK

3 P.

e)

$$F = p \cdot A \quad A = \frac{F}{p}$$



$$\frac{3000}{6.0 \cdot 10^5} = 0.005 \text{ m}^2 = A_1$$

$$A_1 + A_2 = 0.005 + \frac{1}{4} \pi 0.05^2 = 0.0069 \text{ m}^2$$

$$D = \sqrt{\frac{A}{\frac{1}{4} \pi}} = \sqrt{\frac{0.0069}{\frac{1}{4} \pi}} = 94 \text{ mm}$$

4p

Opgave 2 - Pneumatisch schakelen

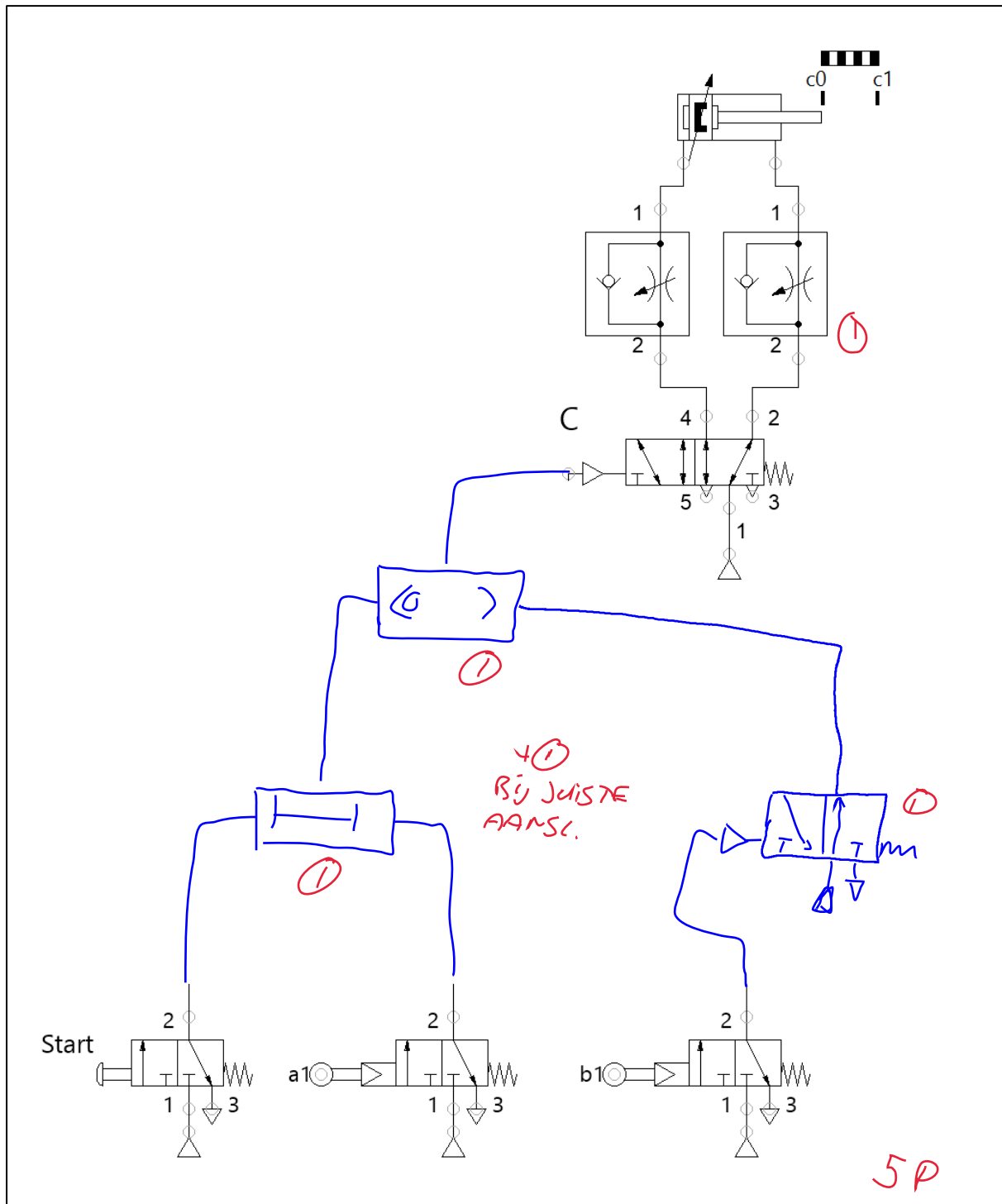
a)

$$C = d? - \text{START} + \overline{h_1}$$

① PER Font

3p

b)



c)

b1	start	a1	C
0	0	0	/
0	0	1	/
0	1	0	/
0	1	1	/
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

① PER Four
2p

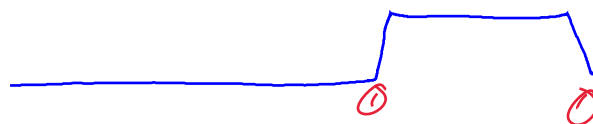
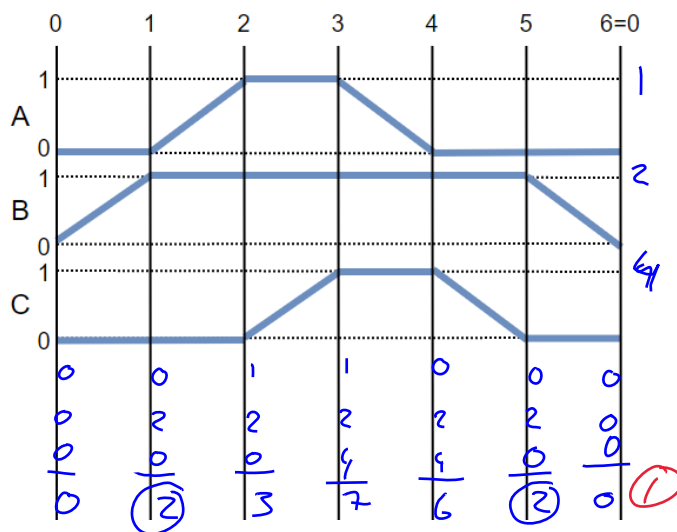
Opgave 3 – Volgordebesturing

a)

$B^+, A^+, C^+, A^-, C^-, B^-$

① PER Four
2p

b)

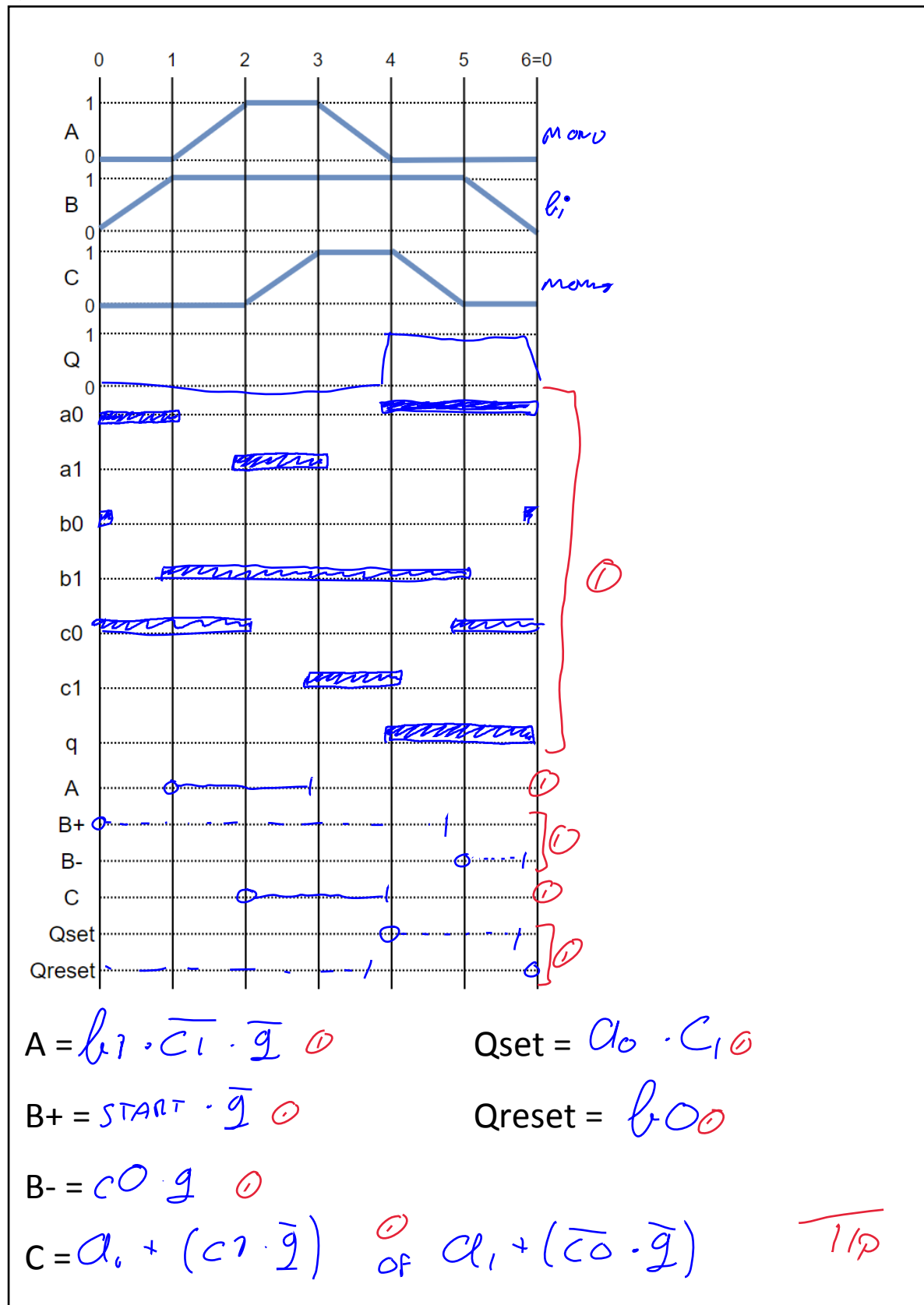


OF $4 = SET$ ①
 $6 = 0 = RESET$ ①

"In het antwoordmodel mag je dat geheugen ook over 2 zetten."

3p

c)



Opgave 4 – Stromingsleer

a)

$$\Delta p = \frac{1}{2} \rho \cdot v^2 \Rightarrow v = \sqrt{\frac{\Delta p}{\frac{1}{2} \rho}}$$

$$1300 \text{ nr} = 13.000 \text{ Pa} \quad (1)$$

$$v = \sqrt{\frac{13000}{\frac{1}{2} \cdot 1000}} = 5,1 \text{ m/s} \quad \text{of} \quad 18,4 \text{ km/h} \quad (1)$$

(1)

/ 3P

b)

$$Re = \frac{\rho \cdot v \cdot d}{\eta} \Rightarrow v = \frac{Re}{\frac{\rho \cdot d}{\eta}} \quad (1)$$

$$v = \frac{2300 \quad (1)}{\frac{1000 \cdot 0,005}{0,001} \quad (1)} = 0,46 \text{ m/s} \quad (1)$$

/ 4P

c)

$$\rho_w = f \cdot \frac{l}{d} \cdot \frac{1}{2} \cdot \rho \cdot v^2$$

$$l = 20 + 2 \cdot 15 \cdot 0,005 = 20,15 \text{ m}$$

$$\rho_w = 0,031 \cdot \frac{20,15}{0,005} \cdot \frac{1}{2} \cdot 1000 \cdot 1^2 = 62465 \text{ Pa}$$

62 kPa
0,62 BAR

3P

Vrije ruimte: